

Erdős Problem #488

EP-488

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let A be a finite set and $B = \{n \geq 1 : a \mid n \text{ for some } a \in A\}$. Is it true that, for every $m > n \geq \max(A)$,
$$\frac{|B \cap [1, m]|}{m} < 2 \frac{|B \cap [1, n]|}{n}?$$